



2354 - Preventing the Slit-Loss Catastrophe Using Flexible, Spatially Resolved Galaxy Models

Cycle: 1, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

NIRSpec MOS is a multiplexing workhorse that can provide high-resolution spectroscopy for hundreds of distant galaxies in a single exposure. Its extreme sensitivity and novel wavelength coverage will provide a completely unprecedented window into the star formation histories and elemental abundances of galaxies in the early universe. However, there is a critical challenge: the micro-shutters are substantially smaller than many targets of interest, meaning that median galaxy at $1 < z < 3$ will have 65% of its light blocked by the shutter. More dangerous perhaps is that due to the intrinsic coupling between stellar populations and galaxy morphology, the light that is measured will unavoidably provide a biased view of a galaxy's stellar populations. Here we propose to solve this problem by developing a tool to build loose, flexible, spatially resolved models of the stellar populations in each galaxy. The model will knit together both resolved and unresolved imaging and aperture spectroscopy into a single unified

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framework by applying the spectrum as an integral constraint over the area of the MOS shutter. The model will be trained on realistic images and spectra from hydrodynamic simulations. The extent to which the shutter spectrum can be generalized to the entire galaxy will be guided by the resolved photometric colors. We will rapidly develop and release this open-source tool to the community in combination with a live demonstration on relevant ERS data in order to maximize the scientific potential of the NIRSpec MOS. Investing early in analysis software developed by the community is an efficient way to ensure the \$10 billion JWST provides the best scientific returns.

OBSERVING DESCRIPTION

No observations are requested.